

Sequential Semantic Segmentation of Road Profiles for Path and Speed Planning

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Abstract—Driving video is available from in-car camera for road detection and collision avoidance. However, consecutive video frames in a large volume have redundant scene coverage during vehicle motion, which hampers real-time perception in autonomous driving. This work utilizes compact road profiles (RP) and motion profiles (MP) to identify path regions and dynamic objects, which drastically reduces video data to a lower dimension and increases sensing rate. To avoid collision in a close range and navigate a vehicle in middle and far ranges, several RP/MPs are scanned continuously from different depths for vehicle path planning. We train deep network to implement semantic segmentation of RP in the spatial-temporal domain, in which we further propose a temporally shifting memory for online testing. It sequentially segments every incoming line without latency by referring to a temporal window. In streaming-mode, our method generates real-time output of road, roadsides, vehicles, pedestrians, etc. at discrete depths for path planning and speed control. We have experimented our method on naturalistic driving videos under various weather and illumination conditions. It reached the highest efficiency with the least amount of data.

Index Terms—Computer vision, autonomous driving, motion profile, road profile, semantic segmentation, temporal sensing, motion detection, path planning, speed control.

I. INTRODUCTION

THE core of autonomous driving systems contains modules of sensing, perception, planning and control. A vehicle receives signals from *sensors* including camera, LiDAR, radar, GPS, etc. The *perception* module can fuse the information [1], [2], [3] to recognize objects and events in the surroundings [4], [5], [6], [7], [8], [9]. Then, a *planning* module plans speed and path ahead [10], [11], [12], and the decisions are transmitted to *control* module to command vehicle actions such as accelerating, braking, and steering [13], [14], [15], [16]. The entire process needs to be in a fast loop to respond to dynamic changes on the road, if a vehicle moves fast.

Driving planning not only requires road shape and depth in path planning, but also object motion for speed planning. A target at far may also cause a collision if it approaches fast.

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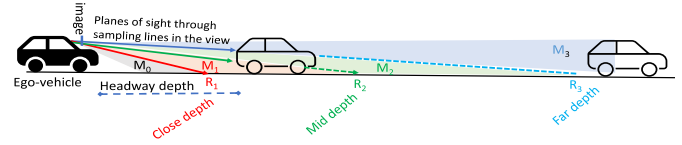


Fig. 1. Side view of the planes of sight through sampling lines. Lines can sense the ground at several depths, as well as objects by these depths at every moment. The sequential semantic segmentation shifts along time axis on video profiles. The network outputs headway for speed planning at several levels (adaptive cruise control). The detected lanes are used in path planning.

Moreover, busy traffic, narrow streets, complex scenes, and fast motion require computationally expensive algorithms in video sensing. However, video may have large view overlaps between consecutive frames. Such data redundancy increases the computing cost in hardware or a latency responding to dangers.

Human drivers selectively perceive the dynamic world from object motion, size, weak depth of stereo, and road structure for fast driving. In this work, instead of analyzing full-view scenes frame-by-frame followed by tracking motion, we improve the sensing efficiency by adopting temporal-to-spatial framework. We only look at sparse horizontal lines in video frames but in dense rate. Taking the advantage that all vehicles are moving forward on a flat plane smoothly, such a sampling line scan horizontal motion and road layout directly into a *video profile* [17], [18], [19] in real time. This reduces a 3D data volume of video into several 2D spatial-temporal images at proper depths. Through a horizontal line in the camera frame as shown in Fig. 1, the *plane of sight* looking ahead in a particular angle will scan the ground once during the driving. It also captures vertical objects closer than the depth constantly [20]. From the estimated vehicle headway and road direction recognized in several *road profiles*, the ego-vehicle can target a proper speed accordingly [10], respond to various events promptly (e.g., cut-ins), and accelerate if the road opens up ahead.

Now, the critical task is to recognize road, vehicles, and other classes robustly in such compact video profiles. In this paper, we challenge the real-time recognition of road profiles for the first time. We employ deep learning semantic segmentation but in the temporal domain. Though the road and motion segmentation in the profiles is different from segmenting normal images [22], [23], [24], we verify that road and motion profiles are learnable from large samples according to local color and global layout of road environment [21].

To minimize sensing latency, traditional patch-mode segmentation is further updated with a sequential scanning-mode to output at every frame in video rate. To efficiently reuse hierarchical temporal memory in shifting a CNN pyramid and avoid repeated computation, a *shifting network* architecture is designed for successive input. The temporal window with length T outputs semantic result of the latest line for vehicle control. By shifting such a window line-by-line over video profiles, we generate output faster than video rate.

We have conducted experiments on videos of naturalistic driving under various weather and illumination conditions. The results show the robustness of our semantic segmentation to illumination changes. Besides, the comparisons to other state-of-the-art methods confirm the advantage of our method for real-time autonomous driving.

The main contributions of this paper are summarized as:

- 1) For the first time confirmed that the temporal-to-spatial road/motion profiles are learnable at pixel level for perception by deep learning semantic segmentation.
- 2) We propose an inference model, Temporal shifting memory (TSM), to avoid data overlapping in sequential semantic segmentation, which is significant to the applications on edge devices.
- 3) The semantic segmentation of RP/MP is further extended to multi-depths for the path/speed planning according to the sensed headway and lane position.
- 4) We build an initial dataset of road and motion profiles with semantic labels from long HD driving videos. The dataset is published with this paper.

The segmentation of spatially sparse but temporally dense scenes will bring following impacts to autonomous driving:

- The sensing will speed up significantly. Thus the control cycle will be as fast as video rate. The path and speed planning can respond to sudden events with zero-latency.
- The temporal scanning avoids heavy shape recognition of sophistic scenes, and it also obtains motion directly in video profiles, as the original goal of driving is to pass a space safely rather than investigating every object.
- The data size to process is much smaller than the full image and video. The approach is thus suitable for light edge devices in driving.

II. RELATED WORKS

A. Driving Video Profiles

The *video profile* [17] has been proposed to record road scenes as vehicle moves along a smooth path. For various applications, a *road profile* (RP) and a *motion profile* (MP) are generated from a driving video. The road profile has been used in the data mining of road appearances under various weather and illumination conditions [19], [25], [26], as it can record long road scenes without repeating in a compact file. The motion profile has been used in pedestrian detection from walking traces [20], [27], computing TTC [28], and filtering vehicle interaction [29]. This paper is devoted to the full segmentation of road profiles necessary for path and speed planning. Based on real-time sensing of road, vehicle, off-road,

pedestrian, obstacle, and then the headway, path and speed can be planned for ego-vehicles.

B. Semantic Segmentation on Spatial Domain

Deep learning methods have demonstrated convincing accuracy in object detection and segmentation. Since the fully convolutional network (FCN) [22], image classification with CNN has been extended to pixel-wise segmentation. Many works [23], [24], [30], [31], [32], [60] have improved network structure for a better segmentation accuracy. They are mainly divided into categories: (i) Embedded with basic encoder-decoder module, re-connect local-global through residual or recurrent molecules; (ii) Short-cut the layers for the sake of efficiency while maintain prediction accuracy. These works provide hint to our segmentation of road profiles in the temporal domain. Temporal association in the streaming profiles should be learned as well in our framework.

Advanced: Recently methods aimed at enhancing global context by adopting attention technique [61] to query temporal relation from past frames [62], [63]. The cost computation makes them difficult for full-view video semantic segmentation in real-time. Some of them rely on complex concatenations of contexture in space or refine loss function for optimization. Generally, the basic idea intends to design a model through heavy computation and memory, which is not application to edge devices for real-time inference in driving.

C. Deep Learning in Temporal Domain

1) *Segmenting Frames Along With Motion Feature (2D+t)*: Many developments have been competing video segmentation. Frames can be semantic-segmented in 2D space with a cost of estimating between-frame dependency from optical flow (1/2-D feature) for improving the accuracy [33], [34]. For a longer term motion, LSTM+CNN methods [38], [39], [40] and recurrent cells [43] interpret temporal changes based on spatial entities extracted by CNN. Such methods predict results at the cost of heavy computation, which is limited to compact instances and their trajectories, rather than a distribution map. Different from such tracking and matching of entities across frames, the segmentation method in [62] shares latent values across frames at different network layers in order to maintain the continuity at semantic levels, though the ad-hoc links from past frames are limited to several frames due to the complication of channel concatenation.

Our video profiles use the least input data, e.g., $6m/720th$ of HD video and m is the number of sampled profiles. Such sparsity in spatial domain is compensated from a dense temporal resolution in recognition and motion detection. The temporal domain of profile contains not only motion but also spatial layout from its acquisition. We need a stronger deep network to learn both motion and shape in the temporal domain.

2) *Video Volume Semantic Segmentation (2D×t)*: Video contains scene trajectories as tubes in the video volume. By treating shape and motion equally, 3D CNN [35], [36], [37], [41], [42] directly encodes and decodes video clips in end-to-end fashion. Due to the memory demand when a 3D

video block is involved, the method has only used short clips and is done off-line. In such a batch-mode processing, only frames in the same clip/block share temporal association back and forth at different levels of abstraction. Also, the method has a latency in output in collecting necessary frames into a patch, which is unsuitable for streaming video input.

For driving, vehicles move 22m/sec at speed of 80km/hr. The segmentation needs a fast rate to respond to dangerous events. Our work is the first one to perform semantic segmentation of a space in the temporal domain. It adopts a scanning-mode or shift-mode on each frame (1 line) as in 2D+t method to yield output without latency. The applied CNN in temporal encoding has multi-level and multi-span referencing to history frames as 2D×t method does. We further design a novel architecture called TSM to avoid overlapped data computation in consecutive frames.

In the following, the paper will introduce video profiles and their use for sensing and driving in Sec. III. Section IV explores the critical deep learning and testing of road profiles. Section V proposes the new network without redundant computation for fast response. Section VI shows experiments and evaluations, followed with the discussion of speed/path planning in autonomous driving.

III. VIDEO PROFILES FOR SENSING

A. Dynamic Projection From Line Scanning

Driving video has a large volume of data (2.6GB/hr for HD video) if it is processed on every frame, while autonomous driving requires fast processing for early response to dynamic events. A delayed computation on large data affects the subsequent path planning and vehicle control. Thus, we employ the video profiles of driving scenes to capture non-redundant road and motion traces directly.

A forward camera is set at windshield. The projection of horizon is specified first at h in the view. A calibration can find camera tilt and pan with respect to the vehicle heading direction from the Focus of Expansion (FOE) and camera roll with respect to horizon, when ego-vehicle moves straight shortly [44]. Below the horizon, multiple horizontal lines $l_1, \dots, l_m < h$ are set at different heights to scan the ground at close, mid, and far depths (three for illustration here). Through these sampling lines and the camera focal point, slanted *planes of sight* cut the space ahead on objects and road (Fig. 1) as follows:

- Objects such as vehicles, poles, trees, pedestrians, etc. with a height (e.g., above curb) will be cut by a sampling line if it is closer than the corresponding depth of the line on the ground. Such an object is sampled at every moment.
- Ground such as road, lane mark, curb, and off-road lawn lower than bumper height are scanned once by a sampling line, and by multiple lines consecutively at their depths.

The geometry of scanning is further displayed in 3D in Fig. 2, where sampling lines intersect the ground surface for road positions, and intersect with objects for avoidance. Multiple planes of sight may cut a close object at different heights.

Given the camera capturing scenes at video rate (30fps) and a sampling line set according to a monitoring depth,

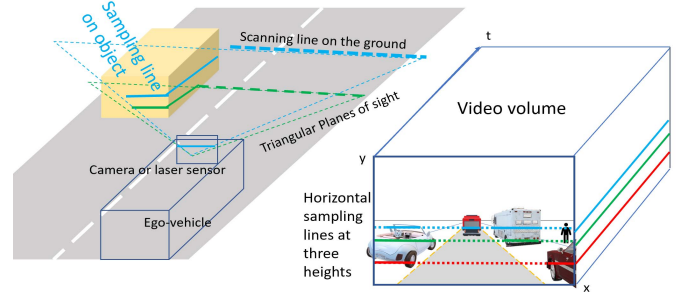


Fig. 2. Line scanning on road and objects. (left) Planes of sight through lines (blue and green) intersecting with the ground (dash lines) and objects (solid lines), respectively. (right) Camera frame with sampling lines l_i (red, green, and blue). Objects are constantly sampled by one or multiple lines.

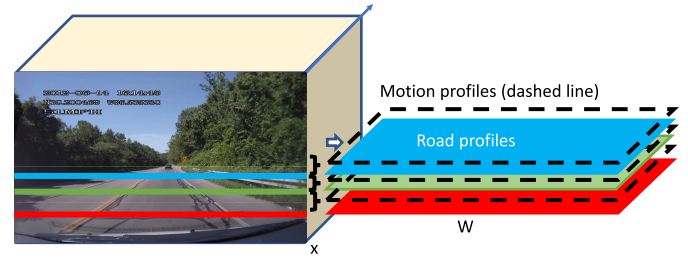


Fig. 3. A road profile (RP) is accumulated from a scanning line and an associated motion profile (MP) is averaged in a belt between the scanning lines. Here three horizontal lines fixed at specific heights in video frame capture profiles from far, mid, and close ranges. In MPs, scenes on the ground are blurred while vertical objects and vehicles are kept sharp as traces.

Z , the plane of sight reaches depth on the ground, and scans vertical objects closer than Z at every moment. We illustrate three depths to generate video profiles for collision avoidance at close range, path planning at middle range, and road navigation at far range.

As the pixels on line l are collected from consecutive frames, a spatial-temporal image $R(x, t)$ called *road profile image* (RP) can be listed up to the latest frame (Fig. 3). The temporal connection of 1D scene reflects a spatial layout. The image memory captures lanes and off-road areas without redundancy. It also covers any other vehicle constantly up to depth Z . If we can identify vehicles and lanes promptly on the current sampling line, we can plan a vehicle path instantly. If a car moves in 54km/h, the sampling line advances half meter per frame in video. In the generated road profiles, the road may be waved mildly due to the camera rolling on uneven road slope. The road or lane may be bended largely due to vehicle yaw change in turning and lane changing (Fig. 4).

One-dimension temporal data seems not sufficient for spatial recognition, but the consecutive collection of lines forms the spatial-temporal image, i.e., a deformed parallel-perspective projection [18] capturing road once. It shows road layout and continuous object motion for shape understanding. Static 3D objects are also mapped to the RP with the temporal scale adjusted by ego-vehicle speed, and slight waving due to camera rolling. With such known deformation, road layout and object structure are recognizable.

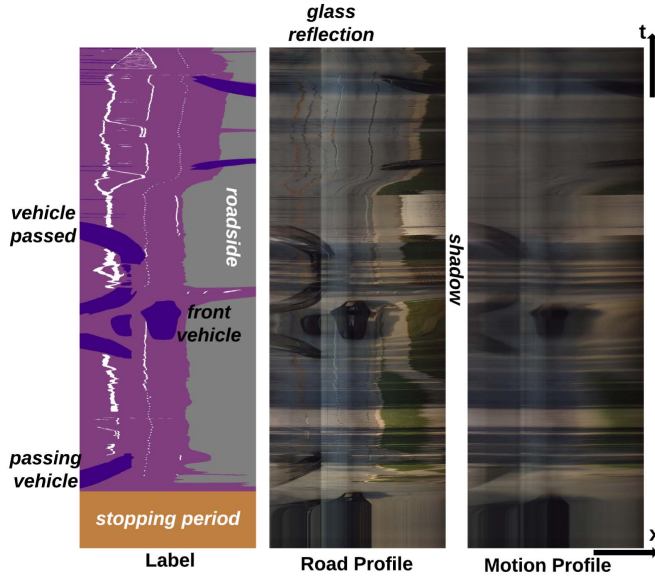


Fig. 4. Road profile, motion profile, and label image from mid-range in complex driving scenes. Ego-vehicle stopping is visible in the road profile. We label semantic classes from material color and motion styles in RP. The MP blurs out road surface but leaves vehicle traces. Instantaneous illumination changes cause horizontal strips when vehicle/camera went into shadows or under bridges. Windshield glass dirty and reflection are at fixed position and form pure vertical lines. The time axis is upward.

B. Motion Profile Image Generation for Vertical Objects

In addition to each RP, we set a horizontal belt between two sampling lines in the frame to compute pixel averages vertically. The averaged pixel arrays from consecutive frames are copied to a memory $M(x, t)$ called *motion profile* (MP) [28] in the same coordinates as RP as shown in Figs. 3 and 4. This averaging blurs non-vertical features such as road edges, while leaves vertical rims of vehicles and objects sharp enough in the MP. The traces in MP thus indicate vehicle existence, and trace width further tells the vehicle depth approximately. For the motion profile $M_m(x, t)$ obtained from the zone above l_m to the horizon in the frame, the traces are from vehicles and background scenes only, which show background expansion, and front vehicles with relatively stable positions. This also gives a clue to identify vehicles from obstacles [45].

Although the scanning effect of the RP is not maintained on a dynamic vehicle, the intersection of scanning plane with vehicle surfaces records the vehicle trajectory (Fig. 4). The surrounding vehicles thus can be recognized from their motion traces such as passing, passed, cut-in, crossing, and approaching [21], [28], rather than from their shapes and types. Vehicles have more variations in shapes than regulated motion on roads. Therefore, our method can directly acquire the motion necessary for driving and skip the cost object recognition, since driving is to pass a free space safely.

Inference of relative motion for interaction. Vertical features such as trees, poles, and vehicle rims are repeatedly captured by lines. Object traces are hyperbola curves in the RP, if ego-vehicle runs straight in a constant speed [45]. Surrounding vehicles have their trajectories in the MP with certain patterns as interaction [29]. Temporally, if the ego-vehicle

has a yaw change (turning), all the road structures move in the inverse direction in RP/MPs. If ego-vehicle stops, background scenes appear as pure parallel lines along the time axis in RP/MP. Passing vehicles at next lanes leave inward trajectories, while vehicles being passed have outward traces. A front vehicle has its trace appearing in RP/MP, and the trace width is changed according to headway. Compared to an RP, MP is more stable even if ego-vehicle shakes slightly on road surface and waves on a hill road. RP provides static road, while MP reveals more of other vehicles.

C. Multi-Level Path Planning and Speed Control

As a vehicle moving on a plane, we scan scenes at several depths corresponding to vehicle speed. The number of depths to monitor, i.e., discrete levels of headway, is selected according to the designed speed levels in the adaptive cruise control for a type of vehicles. For speed levels, V_i , $i = 0, 1 \dots m < 10$ from stop ($V_0 = 0$) to speed limit (e.g., 100km/h), their *braking distances* Z_i are determined for a type of vehicles from tire friction rate and gravity. We set scanning lines l_i at depth Z_i to monitor vehicle's free space ahead for adapting vehicle speed accordingly. Multiple road profiles, $R_i(x, t)$, are then obtained for road or vehicle recognition. A short memory restores the ground between these depths during vehicle motion. A far road profile contains more vehicle traces reflecting traffic flow than road surface, while a close road profile mainly contains lanes on the road. For progressive driving scenes from far to close, an uncertain road area at far range will be refined at mid depth. Moreover, a vertical object closer than Z_i will be scanned by all lines $l_i \sim l_m$ at every moment (Fig. 2) and appear in profiles $R_i \sim R_m$ constantly. If an object is in the driving lane, the headway space, D_h , can be obtained up to Z_{i-1} where R_i is the closest road profile that detects the object. This yields a braking within the headway, while finding a front vehicle in a far headway will slow down ego-vehicle to a lower and safer speed. Human drivers and Adaptive Cruise Control (ACC) have adopted this headway method in speed control.

Laterally, a lane detected at several depths provides the vehicle path. The traffic and road direction can be sensed from far depth profile near the horizon. The middle range profile is used for path planning and finding vehicle interaction, while an urgent cut-in can be sensed at close range in a lower profile.

An HD video yields road profiles with $m/720^{\text{th}}$ (image height is 720 pixels) of the video data size but observe road at m depths. Along with MPs, this significant data reduction to $2m/720^{\text{th}}$ allows a fast processing for event responding at the earliest moment and path planning at every frame. It is always possible to revisit the video frame when a pixel-line at a specific depth has an uncertain result. The location triggered in the road profile provides candidate positions of bounding boxes for reexamining the frame. Therefore, road profiles with compact data size have not lost critical information in normal video frames. Figure 5 shows the relation between a local frame and a profile in the video volume for switching between them when necessary for tasks such as starting and sharp turning.

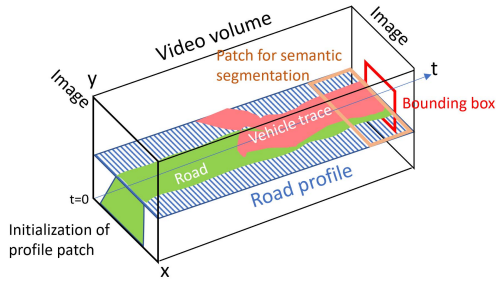


Fig. 5. Uncertain recognition result at the latest input of RP (yellow) can be verified further in the video frame further with bounding boxes (red).

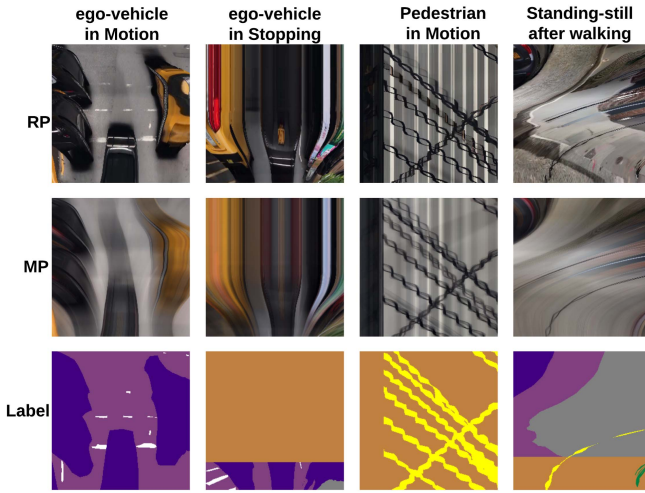


Fig. 6. Various classes of objects and motion. Compared to RP, MP captures object movement by smoothening details in background, which is effective in distinguishing vehicle/pedestrian motion.

IV. DEEP LEARNING OF ROAD/MOTION PROFILES

A. Information in Road and Motion Profiles for Learning

The road profiles have reduced data in a dimension, but they are possibly deformed by the ego-vehicle motion. The motion profile records the motion of surrounding vehicles. These bring challenges in interpreting video profiles. Nevertheless, we have confirmed that human observers can understand most patterns of road and motion in the profiles according to traffic scenarios as in Fig. 6. Our goal is to automatically recognize the road profiles in different classes occupied in pixel level up to the latest moment in the video. We classify pixels in the road profiles into seven classes as:

- **Road**(mauve color): road surface in temporal space for path planning.
- **Roadsides**(grey): adjacent to the road on two sides including sidewalk, grass, and ground area.
- **Vehicles**(purple): moving or stopped vehicles appearing in profiles as trajectories. They will be recognized for vehicle following, avoidance, and interaction.
- **Lane marks**(white): include yellow and white, solid and dashed lines on the road. They are used in path planning.
- **Vertical obstacles**(green): on roadsides including buildings, poles, trees, barriers, and so forth for avoidance.

- **Pedestrian** (yellow): pedestrians in walking for paying more attention at next moment, while standing-still persons are treated as obstacles to be avoided in path planning.
- **Stopping and sharp turning period**(orange): the periods in road profiles while ego-vehicle is stopping or turning. Though such periods may not have sufficient history for recognizing targets due to limited movement, we can switch to a normal image for a slow starting and turning (Fig. 5).

We employ deep learning-based semantic segmentation that has succeeded in normal images based on color, shape and structure [22], [23], [24], [30], [31], [32]. Our method learns not only features such as color and texture, but also the global association of segments that was hard to code in algorithms such as LSTM so far, because we summarize entities by CNN structure. The video profiles have a simpler layout in the horizontal direction than normal images with high buildings and trees, since the profile looks at road scenes lower than the horizon. However, the motion trajectories in the temporal direction of RP/MP are more waved and varied than in normal images due to dynamic traffic and ego-vehicle motion.

The road appearance in the RP is determined from the surface reflectance. Under different illumination conditions, color changes dramatically [19]. The semantic segmentation will learn results less influenced from shadow, highlight, wet surface, and even snow on the road. Through many experiments after initial trials [21], we finalize the concatenation of RP and MP (R, G, B in each) as six-channel input for the network. The sharp features in MP are from vertical objects. White or yellow lane marks appearing in RP only are distinguishable at first layers of deep network. Road and off-road regions are further associated at higher network layers. Roads have different gray levels while off-road has even larger color variations from green lawn to brown soils. The region identification should be fixed by the latent variables through training. On the other hand, pedestrian leg motion obeys a special chain structure in MP along walking trajectory [27], which is memorized in a local structure as long as a stepping cycle can be covered by a temporal window in the computation [20].

Under smooth motion of ego-vehicles, the temporal accumulation of line data to profiles reflects the continuity and casualty in both road space and object trajectory. The smooth scene association or dependency will compensate the sparsity of line sampling in video frames to make profile recognizable.

B. Training Deep Learning Model in Patch-Mode

To achieve a fast response during vehicle driving, our semantic segmentation algorithm processes one patch of RP/MP at a time to output result incrementally. A patch has the image width and time duration T to include the motion continuity of vehicle, pedestrian leg-motion, and road structure changes. The patch is shifted every Δt frames (finest one frame) along the time axis. The semantic segmentation not only grasps local color and texture by convolution, but also global relations between regions through maximum pooling in the network.

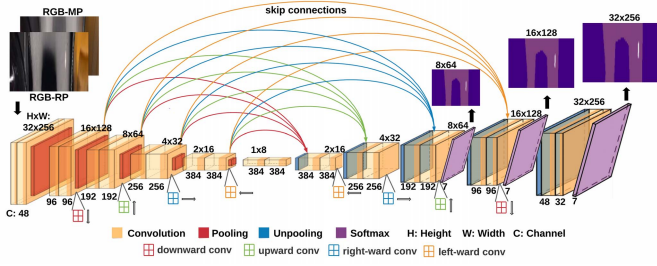


Fig. 7. Encoder-decoder module embedded with fully skip-connections. The raw input consist of 6 channels from RGB-RP and RGB-MP. Each block contains convolution and max-pooling layers. To enhance the continuous segmentation for thin and long objects, we add spatial convolution in several directions on feature maps. The model learns feature maps in multiple scales.

In practice, we set small temporal duration, i.e., T is 32 fr., from one-second video history to perform encoding-decoding network for semantic segmentation. A short T requires less computing time in deep network but may miss some motion continuity. We resize input patch width horizontally from 1280 pixels to 256 pixels, meaning that a road will not have lateral segments (lanes, off-road, etc.) as many as 256. Time duration T is not resized in order to preserve the fine temporal resolution for sensitive vehicle controlling. The training process uses individual patches in Δt interval (e.g., $\Delta t = 4$ frames) to monitor varied positions. In the testing process, however, we shift one line in each advance (i.e., $\Delta t = 1$ frame for zero-latency) to output the latest line for quick responding. This shift-mode causes $T \cdot \Delta t$ lines overlapping between adjacent patches.

The backbone of our training model is an **encoder-decoder** module embedded in a fully convolutional network as shown in Fig. 7. It consists of five encoding and five decoding blocks; each block contains two convolutional layers. To enhance low-level distorted or waved features by deep network, we add skip-connections between every encoding and decoding block [21].

Considering semantic classes including vertical objects and pedestrians in thin and long shape, and pedestrian walking mainly relies on the connected leg-chains in the MP [20], we add an operation of **spatial convolution** [46] within each feature map like *motion blurring*, in addition to the normal layer-wise convolution in the CNN network. In Fig. 6, spatial convolutions in four directions, i.e., downward, upward, rightward, and leftward convolutions, are implemented separately through 2×2 kernel in the network. Following the symmetric encoder-decoder architecture, we add only one such directional convolution in each layer for saving the cost.

For training data, we use a large Naturalistic Driving Video [47]. There has a huge unbalance in pixel amounts across different semantic classes in RP/MP. The specific statistics of datasets per semantic class and weather are given in Fig. 8, in which road and roadside fill the background of road profile while other objects occupy much less areas. This can cause a problem for the training model to capture real semantic features with small data sizes. Also, the cumulative negative loss from background will overwhelm the total loss to degenerate the model. To solve this problem, we replace

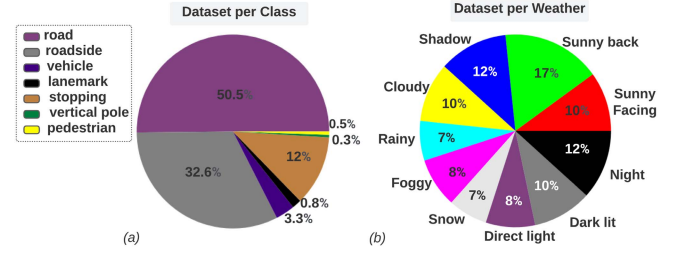


Fig. 8. Sample distributions: (a) Statistics of pixels among different classes sampled from Naturalistic Driving videos. (b) Number of videos for each weather in dataset.

traditional cross-entropy loss with **focal loss** [48] by adding class-sensitive weights to balance the gap among different classes. To make a notation, the estimated probability of a class is p , the focal loss of each feature map is

$$l_{focal} = -\alpha (1 - p)^\gamma \log(p), \quad (1)$$

where α and γ are manually set hyper-parameters.

Finally, to make the model depth-invariant to the training sample, we output the **multiple-scale feature maps** of last three decoding blocks in the network. Given the loss function defined for single feature map in (1), the total loss function is

$$Loss = \sum_{j=1}^m \sum_{i=1}^n (-\alpha (1 - p_{i,j})^\gamma \log(p_{i,j})) \quad (2)$$

where n denotes the number of semantic classes, m denotes the number of feature maps. In practice, we set γ as 2, and α as 1.

V. SCANNING-MODE IN TESTING SEMANTIC SEGMENTATION

A. The Overlaps in Scanning-Mode

In testing phase of segmentation, road/motion profiles are implemented line-by-line with a history window/patch. Thus, there is a data overlapping between consecutive patches when the patch is moved over RP/MP sequentially. This data overlapping is involved hierarchically in the network layers of encoding. For example, as we shift patch *one* line along the time axis for zero-latency output, it has a temporal overlap of $T-1$ lines with adjacent patch; the overlap will then be computed repeatedly in the pyramid structure of CNN network, which causes a large redundancy. Thus, after off-line training in patch-mode, we propose an online testing network in scanning-mode for semantic segmentation during real-time vehicle driving. The new network uses the trained parameters of the network but avoids the repeated computation in overlap part in the network hierarchy. This elaborated mechanism, named *Temporal-Shift Memory (TSM)*, enables testing to output a newest line without accuracy loss so that our segmentation works in scanning-mode in the following.

B. The Periodicity in Network Hierarchy

According to prior works [49], [50], [51], [52] in video deep learning, high-level semantic features will change slower than low-level features across video frames. By setting a kernel size

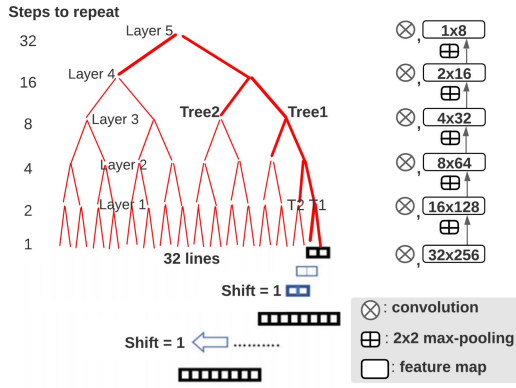


Fig. 9. Periodicity in network hierarchy during its temporal shifting on data to the right. (left) Temporal domain of encoding network. Bold edges indicate the computation for the latest input. (right) network size and structure. Kernel size is 2×2 for convolution and max-pooling, with strides 1 and 2 respectively. There are data overlaps in the network shifting. For example, Subtree T2 repeats the calculation by T1 after two shifts over RP. Subtrees Tree2 repeats Tree1's computation after 8 shifts.

and stride for convolution and max-pooling, temporal shifting of a node at each layer of the network has a period to move on the data covered by another node. As shown in Fig. 9, a subtree over a data section in the pyramid hierarchy was covered by its neighboring sub-tree a period ago during the shifting. By exploiting such periods, we can save and reuse features in a fixed size memory to avoid data re-computing. This has no accuracy loss from trained encoding-decoding model.

As detailed in Fig. 9, assume 2×2 maximum pooling used in the encoding network. A new line entering a network of N layers ($N=5$ here) will affect node values in upper layers until the top of pyramid. The new line in the feature map at each layer fuses with previous ones through 2×2 maximum pooling (marked in bold edges). It can be verified that, at layer k , $k = 0, 1, \dots, 5$, the feature maps have 2^{N-k} temporal lines in the network hierarchy. After 2^k shifts over input RP, lines (subtrees) at layer k move to the adjacent subtrees in the pyramid and the same calculation will be repeated using the learned parameters. In detail, layers 1, 2, 3, 4, and 5 repeat calculation in previous subtrees when RP shifts 2, 4, 8, 16, and 32 steps respectively, while the node values are distinct in between these steps.

C. Memory Structure of TSM

To avoid repetition in data filtering and pooling on the overlapped data of network (Fig. 10a left), we design TSM network to store previously computed node values in time-cells (time instances) for reuse as in Fig. 10a right. A feature map at layer k thus keep a period of time-cells for reuse. At each layer, only the latest nodes (pink) are updated through filtering for the newest input line toward decoding block. Maximum pooling is done between the newly input line and the previously computed line at 2^k cells apart (red edges in Fig. 9a right). Thus, as shown in Fig. 9b, we only update the front surface marked in red when the network moves on the road profile temporally. The computation involves the new input line to the top layer of the network along with the key

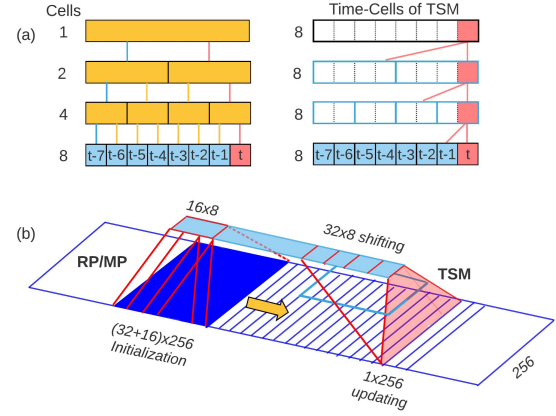


Fig. 10. TSM of 3 layers for illustration of encoding block in the segmentation network. (a) Pyramid network in sideview. The encoding part is depicted. Time-Cells include temporal status of nodes. Update is done only on the newest nodes (marked as pink surface) with key nodes linked by red edges. (b) TSM structure shifting on profiles after initialization in a patch length. Time cells in all layers shift one step backward for next input line.

TABLE I
DATASET OVERVIEW PER TRAINING AND TESTING

	Clips	Frames	Patches	Patch-Size	Shift-Unit
Training	40	40×9000	44,840	32×256×6	8 lines
Testing	20	20×9000	179,360	32×256×6	1 line

time-cells at each layer separated one period in the past. Then, feature maps at all layers of the network shift one position to the past for the incoming line again at next step.

The temporal-shift memory model requires an initial calculation of the network feature maps for later shifting over new inputs. The network needs T lines filled at bottom so as to calculate the top node in TSM. Therefore, the network pyramid skips the first T lines output from the RP/MP. After that, the network only involves one incoming line to compute in scanning-mode.

VI. EXPERIMENTS AND DISCUSSION

A. Dataset and Annotation in Road Profiles

We use naturalistic driving video from both suburban and urban database to generate road profiles and motion profiles. As listed in Table I, a 5-min HD driving clip generates 9000-frame RP and MP. Across ten different weather categories from suburban and urban driving scenes, 40 clips (44,840 patches) are selected from 600 clips for training and additional 20 clips (179,360 locations) are used for testing. For training of patches with length of 32 lines, the temporal interval to cut from RP/MP is 8 lines to ensure sufficient changes between patches. We further set 3 RP/MP at close, mid, and far ranges from each video in the training. In the testing, however, we only shift one line to ensure real-time output with the minimum latency. The testing can be applied up to m RP/MPs ($m = 10$ if necessary) at depths according to the braking distances of ten speeds within speed limit, e.g. 0~80mph.

TABLE II
PIXEL ACCURACY OF SEMANTIC CLASSES PER WEATHER

Pixel Accuracy	Sunny Facing sun	Sunny back sun	Shadow	Cloudy	Rainy	Foggy	Snowy	Direct light	Dark lit	Night	mPA	mIoU
road	0.97	0.99	0.97	0.97	0.87	0.98	0.96	0.99	0.98	0.96	0.96	0.88
roadside	0.97	0.97	0.96	0.98	0.98	0.98	0.95	0.98	0.99	0.95	0.97	0.90
vehicle	0.93	0.97	0.83	0.79	0.94	0.88	0.91	0.98	0.75	0.73	0.87	0.82
lane mark	0.89	0.81	0.78	0.81	0.88	0.85	0.80	0.82	0.86	0.83	0.83	0.80
stopping	0.93	0.91	0.92	0.89	0.88	0.92	0.90	0.99	0.90	0.94	0.92	0.87
vertical object	0.94	0.93	0.91	0.96	0.90	0.97	0.90	0.88	0.94	0.95	0.93	0.89
pedestrian	0.91	0.93	0.94	0.92	0.88	0.90	0.94	0.89	0.91	0.85	0.91	0.84
mPA	0.93	0.93	0.90	0.91	0.90	0.92	0.91	0.93	0.90	0.89		

To annotate pixels in the RPs from three depths, we manually label the whole road profiles of 5 min (9000 frames) with Photoshop in the original resolution (1024 pixels). The regions are painted with the class colors described in Section IV, and we let “stopping period” overwrite other classes except pedestrians. Then, labeled images are spatially down-sampled to 256 pixels horizontally by selecting the median values at region boundaries. We concatenate RGB channels of MP to RP channels for the motion trajectory of vertical objects, which helps vehicle and pedestrian detection.

Based on the illumination in our dataset, we divide driving videos into ten weather categories given in [19], [25] as: Sunny facing the sun, Sunny back to the sun, Shadow, Cloudy, Rainy, Foggy, Snowy, Direct light, Dark lit, and Night. These ten weather categories are all from the natural driving scenes [47] for ablation study to examine the robustness of our model across these illumination conditions.

B. Implementation of Semantic Segmentation

1) *Augmentation*: To avoid risk of destroying temporal information and vehicle motion records, we do not perform data augmentation such as rotation, flip, and whitening.

2) *Kernel Size*: our neural networks adopt small kernel size in convolutional and pooling filters: 1×1 and 2×2 pixels. It has been proved as a good choice [54] that using smaller filters multiple times is better than using a larger filter once.

3) *Neural Network Layers*: we fine-tune all layers by back-propagation through the whole network. For each encoder-decoder module, the down-sampling process reduces the input image to a 32×256 pixel patch and the pyramid network downsizes the patch to 1×8 latent variables with five encoder layers; the up-sampling process in decoding increases the size of feature map from 1×8 back to 32×256 in five blocks, followed by a fully convolutional layer.

4) *Patch Size*: The spatial down-sampling to 256 is to emphasize global segments of road and lanes within the field of view so that they are covered by 8 latent variables. The receptive field covering $1/8^{\text{th}}$ of the view is sufficiently global to make correct decision of lane, track, vehicle, etc., though it is not global enough for determining vehicle stop and turn periods as such actions are generated by ego-vehicle with one latent variable. The temporal span T is set to 32 frames (one second) based on the motion continuity of vehicle and casualty of road changing (also the human perception time in driving).

This duration already covers the most complex leg motion in pedestrian walking cycle (1.26 sec. in average) for pedestrian recognition. Our experiments with a longer history of 64 lines did not show an improved accuracy significantly over 32 lines.

5) *Optimization*: In the experiments, we use RMS as the optimizer to minimize the loss function, and set a decayed learning rate started from 0.0001.

6) *Implementation*: Though the model is trained with deep learning framework of Tensorflow, the testing phase is conducted by applying TSM with pre-trained parameters on video profiles. We program the convolution, max-pooling, ReLU such that TSM does not rely on API. Both training and testing are completed on a local computer with a single GPU of NVIDIA RTX 2700 Super. The training time is 46hr using $1121 \text{ patches} \times 40$ road and motion profiles. The testing time is 0.002 second to generate semantic classes on the latest line from the shifting patch, which is approximately 17 times faster than video rate (30fps). Therefore, ego-vehicle can make an instant response to the input data for path planning.

C. Quantitative Results Evaluated per Weather and Roads

Our experimental results are evaluated from accuracy and efficiency. For accuracy, there are many standard metrics in benchmark datasets for autonomous driving [55] such as Intersection-over-Union (IoU), Pixel Accuracy (PA), Mean Pixel Accuracy (mPA), Mean Intersection over Union (mIoU) and FWIoU. This work adopts per-class PA and IoU for evaluation and weather-based mIoU are also tested.

We have tested our network on multiple weather conditions with numerical results per weather presented in Table II. Figure 11 visualizes the segmentation across different weathers on suburban road. Based on the results, we can find (1) our model is robust to illumination changes even in night driving. Considering the road edges are invisible in normal video frame at night, our segmentation on profiles with temporal association in addition to spatial layout achieves better performance in dim lightness. (2) Under the condition of shadow and rainy, our method filters out their influence from wiper and rain drops. In the road profile for raining in Fig. 11. Rain drops on windshield and water reflection on the ground are troublesome in the scene understanding. Nevertheless, the semantic results successfully segmented the road and off-road.

We also give the numerical result per semantic class in Table II. Our work performs semantic segmentation on

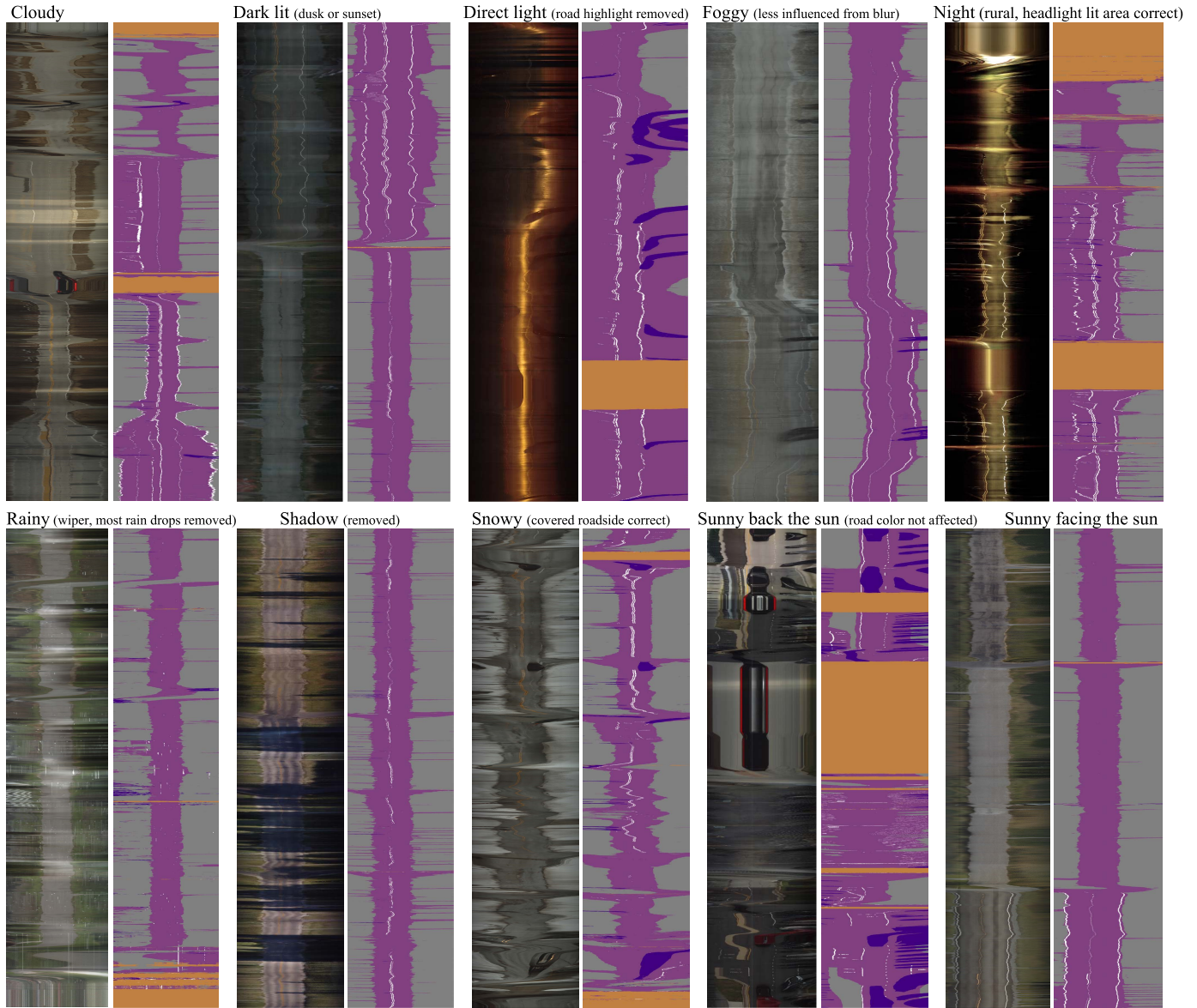


Fig. 11. Semantic segmentation results of road profile (mid-range) in ten weather categories. The time axes are upward. Each sequence lasts for 5min.

both objects and events (vehicle occupancy) in the temporal domain. When the driving vehicle encounters turning or stopping, the semantic features overwrite objects and backgrounds. The detection results of ego-vehicle stopping in Table II implies a good performance in finding spatial association.

In addition to the detection accuracy within each class, we analyze the inter-class association. The pre-defined classes can be divided into three categories: the ground, objects, and event. Table III illustrates semantic pixels classified correctly or miss-classified as similar classes. For instance, road is easy to be classified as roadside, and likewise inversely. Parked vehicles are classified as vertical objects, which is correct. These experimental results can be employed in advance to improve the design of penalty item in optimization. They provide further details on inter-class distances in discriminatory space.

We have tested more complex driving videos of New York City that suffer from frequent stopping of surrounding vehicle,

TABLE III
CROSS-TABLE OF POSITIVE PIXELS FOR INTER-CLASS

True + (%)	rd	rs	ve	lm	st	vo	ped
road (rd)	96	3	0.5	0.1	0.4	0	0
roadside (rs)	2	97	0.2	0.1	0.7	0	0
vehicle (ve)	4	5	87	0.1	3	0.7	0.2
lanemark (lm)	10	3	2	83	1.8	0	0.2
stopping (st)	3.8	2.7	1.1	0.2	92	0.1	0.1
vertical object (vo)	1	0.9	3	0.1	1	93	1
Pedestrian (ped)	1.8	0.5	3.2	0.3	3	0.2	91

which results in Fig. 12. We can observe that the ego-vehicle stopping periods are noisy. The visual cue to judge the stop is the absolute static scenes, which are frequently occluded by pedestrian crowds and busy traffic. Although a single across-field latent node at higher network layer can summarize the stop periods better, we leave them as pixels in result.

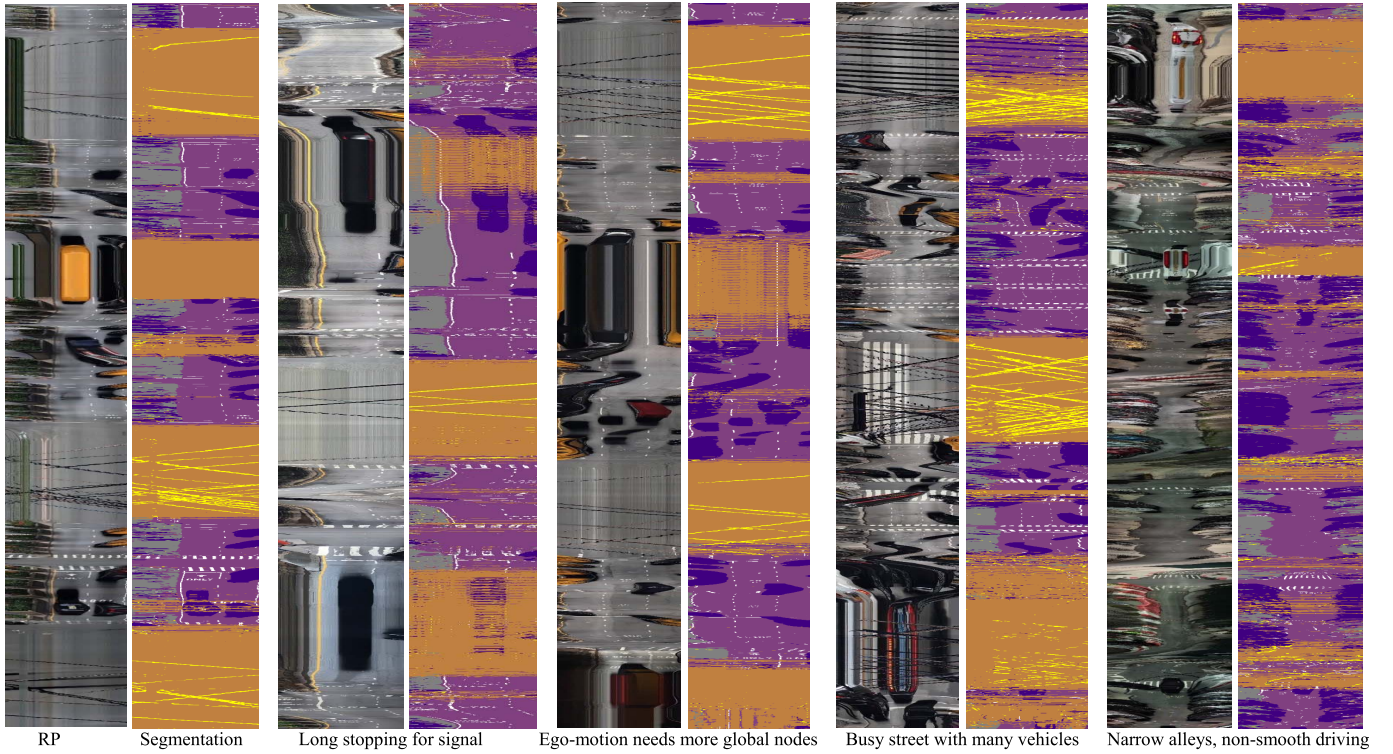


Fig. 12. Semantic segmentation (SS) of road profile (mid-range) from New York City (time axes upward). The road area shows the movable area on road, and many surrounding vehicles detected as traces. The gradient of pedestrian or vehicle traces reflects their relative moving velocity and direction.

TABLE IV
ABLATION STUDY OF DIFFERENT METHODS IN PATCH-MODE TESTING

Methods	base	skip-conn.	spatial	multi-scale	focal	rp+mp
road	0.91	0.94	0.94	0.95	0.95	0.92
roadside	0.93	0.96	0.95	0.94	0.93	0.94
vehicle	0.83	0.87	0.86	0.84	0.85	0.84
lane mark	0.80	0.83	0.83	0.84	0.85	0.82
stopping	0.88	0.91	0.89	0.90	0.87	0.90
vertical obj.	0.89	0.90	0.92	0.91	0.93	0.91
pedestrian	0.85	0.91	0.94	0.87	0.90	0.89
<i>mPA</i>	0.87	0.90	0.90	0.89	0.91	0.89
Time (ms/patch)	14	15	18	19	16	15

Furthermore, different from individual video frames, our profiles also yield the relative velocity of objects from the gradient in the segmented regions [29].

D. Ablation Study on Network and Comparison to SOTA

To evaluate the approach and demonstrate the effectiveness of network component, we study the accuracy under real time setting. We found that adding/deducting an auxiliary technique is a trade-off between accuracy and efficiency. Here we set a naïve encoder-decoder architecture as the baseline working for RP. We explore the differences after adding auxiliary techniques [21], [23], [46], [48]. Summarized in Table IV, it is observed that skip-connection yields more competitive results compared to baseline model. Then, spatial convolution at four directions enhances the detection for pedestrian

and vertical objects, though it increases the inference time. Further, we compare our effectiveness by replacing traditional cross-entropy loss function with the focal loss function, which balanced the data samples between classes. Compared to the baseline, the accuracies in detecting vertical object, lane mark and pedestrian are improved much more than other classes. Finally, combining RP and MP as the raw input can increase the accuracy of segmentation, especially for pedestrians and vehicles. Hence, we found that additional techniques improved baseline with a bit more degree.

For the comparison with other state-of-the-art methods of semantic segmentation [23], [30], [31], [32] that were applied to normal images so far, we run such algorithms and networks on our temporal video profiles. Based on the same dataset of road profiles and motion profiles trained by these models on our local machine, we test and evaluate the sequential semantic segmentation by in scanning mode. Given the temporal-shift as $\Delta t=1$ line, we summarized the results in Table V. Our network, which is a sort of combination of all other methods with TSM, outperforms others in accuracy and efficiency. Particularly, a great improvement in efficiency has been achieved in segmenting latest input line; ours is around 6~9 times faster than other state-of-the-art methods, including complex FRRN [32] with multiple residual modules, and light-weighted BiSegNet [59] with short-cut in up-sampling. This is due to the TSM avoiding the repeated computation in network hierarchy in scanning-mode.

Our work is the first to implement semantic segmentation on temporal domain of driving video in 30fps. So far, there is no other public labeling dataset for semantic segmentation with

TABLE V

COMPARISON TO STATE-OF-THE-ART FOR SEMANTIC SEGMENTATION.
SPEED FOR ZERO-LATENCY OUTPUT IS COUNTED
BY LINES/SECOND IN THE TEST

	SegNet	PSPnet	Deeplab	Frna	Frnb	BiSenet	TSM
mIoU	0.81	0.84	0.85	0.88	0.89	0.90	0.90
line/sec	80	74	71	62	53	178	488
Memory	6.7M	12.8M	14.8M	13.1M	13.9M	11M	0.7M

Acc.=accuracy; Eff.= efficiency in testing measured by frame rate; In testing, temporal-shift unit $\Delta t=1$ line.

such a dense frame rate in the profiles. Although those 2D deep learning [22], [23], [30], [31], [32] achieved convincing accuracy in normal frames (e.g., FrnA reports mIoU of 65.7% in Cityscapes [56]), ours concludes that both dynamic motion and static road scenes can be learned at pixel level in a reduced dimension of video.

To verify the effectiveness of TSM in testing phase, we report the testing rate of different training models with and without applying TSM. The performance of six methods is recorded in Table V. Based on the same volume of training parameters, TSM is significantly faster than other backbone networks running in shifting-mode. Note that it achieves the same accuracy as the training models. Thus, our method reaches faster speed compared to all the other methods all in shift-mode. We open the initial dataset of video profiles including 60×3 manually labeled road profiles and associated motion profiles at three depths [65]. This will provide the common base for further study and development of this representation.

E. Quantitative Results per Depth for Path/Speed Planning

For a driving video, three RPs and their MPs are extracted from close, mid and far ranges. In Fig. 13, we have tested video profiles at each depth over different classes. Compared to semantic results from close and mid-range, the accuracy at far is lower for the reasons of low object resolution in the profile and high traffic density. In the real path planning and vehicle control, the driving scenes are scanned from far to close progressively by different lines. The scenes segmented in farther distance may appear at closer depth again as vehicle moves forward. Multiple lines are sufficient to cover the whole scenes for frequent path and motion planning. We apply the result at mid-range for lateral path planning along with the road direction from far profile, which is further guided by GPS and HD map based navigation. Second, the depth of a vertical object between two sampling lines can be approximated from its width in MP. The TTC to the object can also be obtained instantly [28]. Third, as long as we detect a vertical object in a series of RPs, the closest one thus gives the *headway* length, based on which ego-vehicle will slow down. A targeted speed should have its *braking distance* shorter than the headway. In this way, a motion planning module is designed as ACC to perform vehicle following task.

Based on instantaneous segmentation results of road profiles from sequential input, ego-vehicle obtains obstacles (e.g., pedestrians or other vehicles) at known distances.

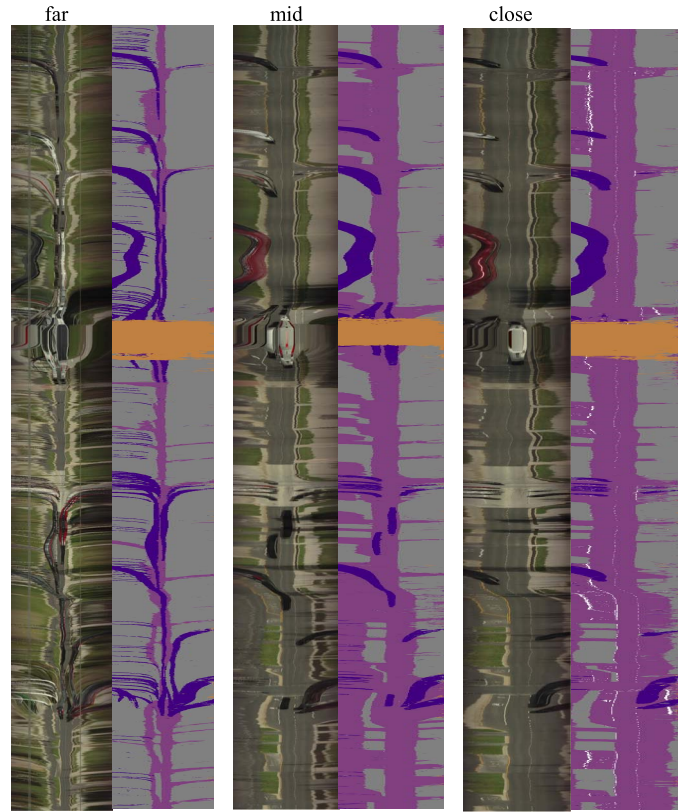


Fig. 13. Semantic segmentation results of road profiles at far, mid and close ranges in the same video. Driving lane is at the center of road profiles. Time axis upward for 5 minutes (9000 frames). The horizontal resolution is 1280 pixels. As driving scenes are observed gradually from far to close, an ambiguous road can be refined by path at closer ranges.

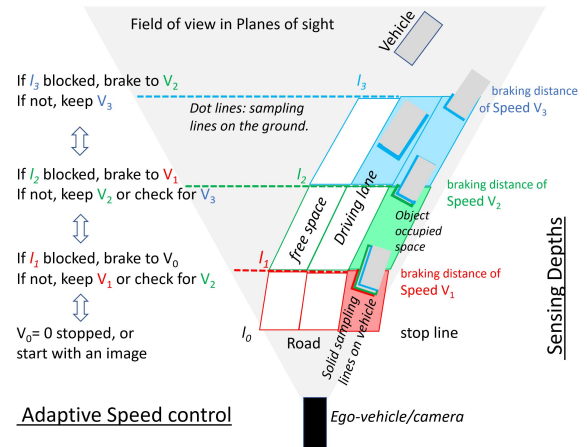


Fig. 14. Block-wise headway space for speed planning. Free space is obtained from the planes of sight intersecting the ground. Object vertical surfaces are viewed by at least one plane. Three scanning lines are at depths marked in color.

As depicted in Fig. 14, the closest road profile yields free headway space, as objects are sampled at every moment by at least one sampling line in the visible range. Meanwhile, road space is intersected by planes of sight as free space. With the lines captured at far, mid, and close depths, vehicle path planning obtains free area at denser video rate (30fps) as shown in Fig. 15.

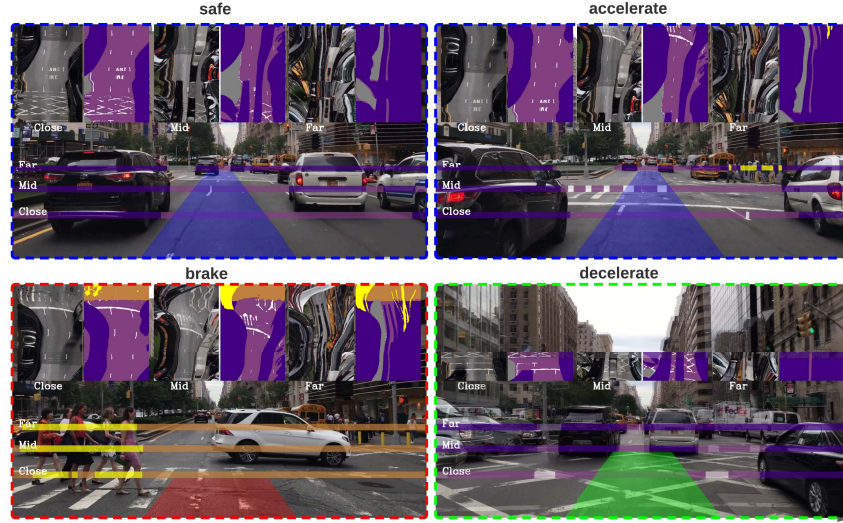


Fig. 15. Speed planning at three-levels according to road profile segmentation for NYC videos. The emergency degree decreases from close to far depths. (top-left) Safe to keep current speed as a vehicle detected only in far range. (top-right) Accelerating if no objects detected at driving lane. (lower-left) Braking when pedestrians are detected at close range. (lower-right) Decelerating when a vehicle appears at middle range. The top half of frames is the real-time semantic segmentation of profiles.

With m lines to produce RPs and MPs, we can monitor road and objects at m depths. From the closest line l_i where an object is detected in $R_i(x, t)$, the headway D_h is located up to l_{i-1} . Based on the criterion that braking distance should be kept shorter than the headway in safe driving, i.e.,

$$D_h > B(V) \quad (3)$$

where ego-vehicle targets at a lower speed V preparing for full stop [10]. Similar as the adaptive cruise control (ACC) based on radar, we set 10 target speeds V_i up to a limit. With 10 sampling lines are set at braking distances $B(V_i)$, we can detect a headway on driving lane. Then, a lower speed V_{i-1} is targeted to ensure $B(V_{i-1}) < D_h$. Deceleration can be cancelled if the headway extends again. On the contrary, if the closer line, l_{i-1} , also detects the vehicle after reaching V_{i-1} , a further deceleration to even slower speed is triggered. Figure 16 is the segmentation from 10 lines to provide smoother speed transition and path.

For discussion, our semantic segmentation on road and traffic in the road profiles is less dependent on object appearance but more on the scene motion, position, and color, under the constraint that vehicles move on flat road and lane regularly [29]. Therefore, a far road profile may not have sufficient resolution to separate road and vehicles as focused recognition in video, if traffic is crowded there. The far profile may obtain irrelevant scenes occasionally from background or even sky, if the road has already turned away at a closer range such as sharply curved road and T-junctions, or ego-vehicle is near a hilltop. Fortunately, all these cases require a slower vehicle motion or stop, for which mid-close road profiles (even static image) can sense the road layout. Separated from the semantic segmentation of designed classes, other collision danger such as bicycles and motorcycles can be measured from TTC [28] or depth by LiDAR. As for the segmentation accuracy, the stopping/turning detection is noisy laterally, which is not as

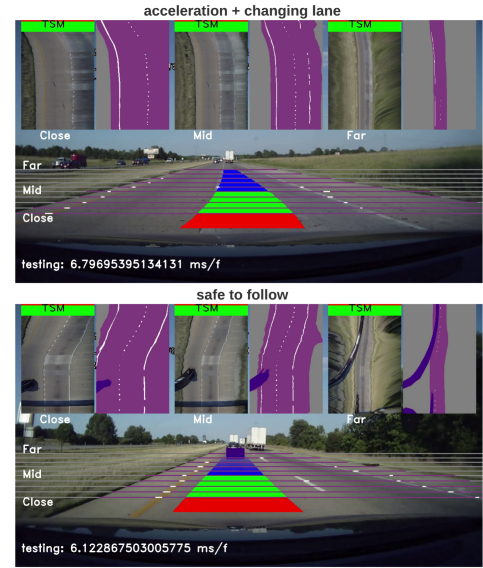


Fig. 16. Path planning using 10 road profiles. The more the sampling lines set in depth, the smoother speed transition can be planned. Non-obstacles detected from close to far range in vehicle acceleration (lower) or lane changing (upper). The TSM network applied regions are displayed in green at top of road profiles above.

uniformed as a non-deep-learning method [10]. This is because of the eight top latent variables in the encoding module running in parallel, are not global enough to conclude an ego-vehicle status. More layers fusing them to one node is necessary.

In general, above sensing and control framework is efficient for fast moving vehicle on road, where the line scanning acquires large enough space in one second for robust RP segmentation. For slow vehicle motion in short distance such as parking, however, traditional image analysis is matured for finding free space and avoiding collision. Our algorithm

collects first T lines into road profiles for semantic segmentation. At infrequent moments when scenes are uncertain in RP segmentation, we can examine their appearance in the frames [57], [58] (Fig. 5) to guide slow vehicle motion. For starting, stopping, and back-up (in parking), an image is needed for initial segmentation because a slow vehicle has not scanned a large area. For sharp turn at a crossing, vehicle/camera rotation captures scene transition in the view as parallel motion traces in the profiles, which is difficult to segment road and traffic. Only the closest RP catches curb or lane mark shortly, whereas the scanning lines of mid-far RPs are already beyond the turning road after the vehicle arrived at crossing. We have promptly segmented the turning moment in RPs for image input. Road segmentation is triggered in the image for turning. Meanwhile, the close RP can still follow road edges for feedback control, until far-mid roads shift back into RPs for road following. Overall, such switching between temporal and spatial domains can increase the recognition efficiency during fast motion and slow maneuver of vehicles, balance different driving tasks, ensure fast responding to dangerous events, and reduce computing resources on vehicles.

VII. CONCLUSION

The main contributions of the work are in the recognition of spatial-temporal data reduced from driving video for motion/path planning of vehicles. By streaming input of video profiles with dense temporal resolution, we infer roads, vehicles, and events for autonomous driving with sequential semantic segmentation, which is the first effort so far on this media representation. Although only 1D data are sampled from video frames, their temporal concatenation in 2D mapping reveal the space and motion trajectories incrementally in driving. To perform semantic segmentation in scanning-mode, we designed a new network called TSM without redundant computation. With the minimum data volume, our network reaches fastest detection rate compared to other methods. The obtained accuracy has convinced the learnability of spatial-temporal profiles, which yields the zero latency. It can plan drivable regions and speed in video rate according to the headway.

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